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Guardians of heritage: women's position in traditional seed systems and agroecology in Zimbabwe

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Abstract

Background Agroecology integrates ecological agricultural production with community self-reliance, deeply rooted in indigenous knowledge. This study investigates how and why women in Zimbabwe contribute to seed preservation in agroecology, and what challenges they face in sustaining traditional seed systems. Using a qualitative approach combining focus group discussions, semi-structured interviews, and participant observation, I examine women's contributions to seed saving, preservation, and exchange, practices vital for sustainable food production and agricultural heritage.

Results The findings reveal four major themes: the cultural significance of seeds; women farmers' specific roles in seed systems; challenges in traditional seed preservation; and implications for food sovereignty and nutrition security. Women safeguarded drought-tolerant crops such as millet, sorghum, and rapoko, which ensured household food availability during climate shocks and 'hunger periods.' Their seed work sustained dietary diversity and nutrition security through protein-rich legumes, nutrient-dense small grains, and multipurpose crops like pumpkins. By blending traditional seeds with selected hybrids, farmers reduced dependency on purchased seed, improved climate resilience, and maintained culturally significant varieties. Seed sovereignty, expressed through these practices, was directly linked to greater household autonomy over planting decisions, reduced vulnerability to market fluctuations, and strengthened local food systems.

Conclusions Recognising and empowering women in agroecology is crucial for promoting equitable and sustainable food systems. The study shows that women's seed stewardship contributes directly to household food security, nutrition, and sovereignty, while sustaining cultural identity, intergenerational knowledge, and biodiversity. Supporting women's roles in traditional seed systems is essential for addressing contemporary and future challenges in sustainable agriculture.

Keywords Agroecology, Food security, Food sovereignty, Gender and agriculture, Indigenous knowledge, Traditional seed systems, Women farmers

Background

In sub-Saharan Africa (SSA), Zimbabwe's agricultural landscape is undergoing significant transformations, particularly due to the Fast Track Land Reform Programme (FTLRP) implemented in 2000 [1]. Among the most affected are smallholder women farmers, who play a pivotal yet often under-recognised role in managing traditional seed systems and advancing agroecological

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practices, such as preserving diverse indigenous varieties, using intercropping to protect soil and moisture, and organizing community seed exchanges [2]. Seeds are foundational to life, they enable harvests, sustain biodiversity, and embody cultural continuity [3, 4]. Globally, farmer-managed seed systems supply 80–90% of seeds in developing countries, making them central to agricultural productivity, biodiversity and resilience [5, 6]. This supports Santeramo's [7] argument that food security research must look beyond yield metrics to include the socio-cultural and institutional contexts that shape agricultural sustainability. In Zimbabwe, farmer seed systems remain the dominant seed source, as evidenced in district-level case studies [8]. In this study, I explore the critical question: How and why do women in Zimbabwe contribute to seed preservation in agroecology, and what challenges do they face in sustaining traditional seed systems?

Agroecology understood here as both a scientific approach and a social movement, integrates ecological principles with agricultural production and highlights the agency of farmers, especially women, in co-producing knowledge and managing biodiversity. At its core, agroecology values biodiversity, traditional knowledge, and local food systems [9, 10]. Traditional knowledge is not only a repository of cultural memory but also a proven survival strategy [11]. In Zimbabwe and other parts of SSA, these skills have enabled farming communities to avert hunger during droughts, navigate pest outbreaks, and adapt crops to shifting climates when formal systems failed to provide timely solutions [12]. In this study, agroecology is understood as the everyday farming practices that accentuate biodiversity, low external inputs, and farmer-led knowledge exchange. Among the communities in Matabeleland North and South, these practices are closely intertwined with traditional seed saving, encompassing the selection, storage, and exchange of diverse local varieties suited to specific soils and climates. For many farmers, these activities are not only agronomic strategies to adapt to droughts, secure household food supplies, and maintain control over production, but also a means of sustaining cultural identity and local autonomy [13]. While participants did not always use the term 'agroecology' explicitly, they consistently described methods such as intercropping, composting, and seed preservation in ways that align with agroecological principles.

In many parts of Africa, women farmers are the primary custodians of this agricultural heritage, particularly in the realm of seed preservation and management [10]. In Zimbabwe, these seed systems include staple crops such as sorghum, millet, rapoko, cowpeas, ground nuts, and many others, which are deeply rooted in cultural and ecological traditions and well adapted to local agroecological conditions. Their conservation and exchange are largely carried out by women, who acquire, preserve, and share seeds using techniques passed down through generations [8]. Their contributions are vital for the maintenance of indigenous seed systems, which in turn are crucial for the food security and nutritional wellbeing of local communities and the empowerment of women [14, 15]. As Vandana Shiva stresses in *Seed Sovereignty, Food Security*, women are at the forefront of defending seed freedom and sovereignty, paving the way to achieving both sustenance and autonomy [16].

These practices are not only agronomic but are also central to household food sovereignty, cultural identity, and nutritional diversity. Recent studies by Onyeaka, Siyanbola [17] and Nchanji, Ageyo [18] argue that supporting women's traditional seed work enhances household food resilience and nutritional outcomes, particularly in marginal agroecological zones such as the one under investigation here. These insights resonate with the findings of this study and inform its focus on food sovereignty; defined here not only as the right to food, but as the capacity of communities to control what they grow, preserve, and eat in ways that are in harmony with their cultural, spiritual, and ecological values [19, 20]. While food security is often assessed through standardized quantitative indicators such as the Food Insecurity Experience Scale [21], such metrics rarely capture the relational, cultural, and empowerment dimensions central to feminist agroecological perspectives. This study builds on such work by viewing food security through a feminist agroecological lens, which sees autonomy, cultural relevance, and gender relational empowerment as core components of food systems, not merely yield or calorie intake [19, 22]. Food security also has demonstrable links to subjective well-being, with cross-country panel evidence showing that improved food security significantly enhances life satisfaction and overall quality of life [23].

However, the sustainability of these seed systems is under mounting pressure [24]. Climate change, hybrid seed promotion, and formal seed certification regimes increasingly threaten the integrity and viability of indigenous seed varieties [25, 26]. In response to these challenges, agroecology women's groups have emerged in SSA communities, providing platforms for women to share knowledge and best practices in sustaining indigenous seed systems [22, 27]. Gendered inequalities, such as unequal access to land, credit, and extension services further constrain women's ability to maintain seed sovereignty (Nyantakyi-Frimpong, 2017). While Non-Governmental Organizations (NGOs) and agroecology groups such as Zimbabwe Smallholder Organic Farmers Forum (ZIMSOFF) provide important support, many women's

efforts remain informal, undervalued, or excluded from agricultural development discourse [27].

This study draws on an intersectional Feminist Political Ecology (FPE) framework, which provides tools to analyse how gender, class, and age intersect with power structures and access to environmental resources [28, 29]. This is complemented by selective insights from ecofeminism, which connects environmental sustainability to gender justice [30]. This intersectional approach allows for a nuanced analysis of how gender, power, and cultural identity shape women's roles in seed systems, while attending to broader institutional and environmental forces. Methodologically, I employ field based qualitative methods, including focus group discussions (FGDs), semi-structured interviews and participant observations with women farmers and community leaders in Zimbabwe's Matabeleland North and South Provinces, specifically in rural communities where women play a significant role in agroecological seed practices. By examining the agency, practices, and challenges faced by women farmers in this context, I seek to contribute to the growing body of literature on gender and agroecology in SSA.

This research is guided by three objectives: identifying the specific practices and knowledge systems that women farmers use in seed preservation; defining the cultural and spiritual significance of seeds in Zimbabwean communities; and analyzing the implications of these practices for food sovereignty and nutrition security. Ultimately, the study aims to provide insights that can inform policy and practice in supporting women's position in agroecology for sustainable agriculture and food security. In the next section I expand on the theoretical perspectives introduced above, clarifying how feminist political ecology and ecofeminist insights inform the analysis of agroecological seed systems, power, and food sovereignty in this context.

Theoretical framework

My research is grounded in Feminist Political Ecology (FPE), which provides the central analytical framework for examining how Zimbabwean women engage with, sustain, and negotiate traditional seed systems within agroecological contexts. FPE offers critical tools to interrogate how gender relations, power structures, and social difference shape access to and control over environmental resources, including seeds and how these dynamics unfold within broader political-economic systems [28, 29] such as that of gender, food production, and property rights. In particular, FPE highlights how seemingly everyday practices like seed saving are embedded in unequal institutional landscapes, household relations, and historically produced patterns of exclusion.

To deepen this analysis, I incorporate an intersectional lens [31] that illuminates how gender intersects with age, marital status, socioeconomic position, and geography to shape differentiated experiences among women farmers. This approach moves beyond a homogenous view of 'women' to reveal how women in female-headed households in remote areas, for example, may experience resource constraints differently than married women in peri-urban zones. These differences are not incidental; they are structurally produced through overlapping systems such as land tenure regimes, access to extension services, cultural expectations, and financial marginalization. Together, FPE and intersectionality provide a framework for understanding how gendered seed work is situated within and shaped by broader configurations of power, identity, and resource governance.

To complement FPE further, I draw selectively on ecofeminist perspectives; particularly those that foreground women's ecological knowledge, biodiversity stewardship, and resistance to corporate seed regimes [16, 30]. Ecofeminism helps surface the relational, affective, and spiritual dimensions of seed practices, especially in relation to cultural meaning, memory, and intergenerational transmission. However, my use of ecofeminism is critical and partial. I reject essentialist claims that women are inherently closer to nature [32]. Instead, I interpret women's association with seed conservation as socially constructed and historically developed, rooted in lived labor, embodied knowledge, and cultural practices passed down through generations [33].

This theoretical combination allows me to analyse traditional seed systems not as static repositories of heritage, but as dynamic, negotiated, and politically charged spaces. FPE reveals how women's seed work can simultaneously be a form of environmental stewardship, cultural continuity, and resistance to institutional marginalization. It also allows for attention to the contradictions and tensions within agroecological practice, such as the dual burden women face in maintaining traditional knowledge while adapting to hybrid seed pressures, donor expectations, and climate volatility. For instance, while NGOs may promote 'climate-smart' framing to gain funding, many women continue to express their seed knowledge through ritual, storytelling, and kinship ties; forms of expression often undervalued by institutions.

In addition, this framework enables critical engagement with dominant food security paradigms. Rather than viewing food security solely through the lens of caloric intake or yield productivity, I adopt a feminist agroecological perspective that redefines food security in terms of food sovereignty, relational autonomy, and cultural relevance. This perspective corresponds with critiques of technocratic and gender-blind food security

models [34], emphasizing that who controls seeds, what is grown, and how knowledge is shared are all deeply political questions. By foregrounding women's seed work, as knowledge holders, family providers, and environmental stewards, this study situates food security not only in terms of availability, but also agency, recognition, and sustainability. Food sovereignty, in this context, is not merely about resisting market dependency, but about relational empowerment rooted in land access, communal decision-making, and intergenerational care. Thus, the framework adopted here interprets seed systems as central to achieving not just resilient food systems, but gender-just agroecological futures.

Throughout the article, this theoretical framework is applied to interpret women's seed practices not only as technical or ecological acts, but as gendered, situated, and political processes. Rather than viewing empowerment as a fixed outcome, I conceptualize it as relational and partial, unfolding through everyday negotiations in fields, homes, and community networks. This framework therefore underpins the analysis in the Results and Discussion sections, enabling a reading of Zimbabwean seed systems as sites of both continuity and change, where cultural identity, agroecological resilience, and gendered power intersect.

Methodology

Research design and approach

In this qualitative study, I explored women's seed-saving practices in Matabeleland North and South, Zimbabwe. Zimbabwe's agricultural sector is largely smallholder-based and rain-fed, making it highly vulnerable to climatic variability and market shocks [35]. While standardized global metrics such as the Food Insecurity Experience Scale (FIES) [21] are valuable for measuring household food access, they do not capture the cultural, spiritual, and relational dimensions of seed work that are central to this study. I therefore chose a qualitative approach to illuminate these embedded practices and meanings, which are often invisible to quantitative measurement tools. Fieldwork took place intermittently between July 2022 and January 2025 in Bubi, Matobo, Nkayi and Plumtree districts. The staggered timing was intentional to capture the full agricultural cycle and observe seed-related practices across different seasonal stages, including seed preparation, planting, harvesting, preservation, storage, and exchange. Conducting field visits at these intervals allowed for richer, longitudinal insights into how women's seed practices adapt to shifting climatic conditions, resource availability, and social dynamics throughout the year.

These areas were selected for three main reasons. First, they represent contrasting agroecological zones within Zimbabwe's semi-arid southwest, allowing examination of how local climate and soil conditions influence seed-saving practices. Second, they encompass significant cultural and linguistic diversity (Ndebele, Kalanga, Shona), enabling the study to capture variations in seed-related traditions, beliefs, and social norms. Third, both provinces have active agroecology networks and established women-led seed initiatives, making them strategic communities for understanding the intersection of gender, agroecology, and seed sovereignty in practice. The Zimbabwe Smallholder Organic Farmers Forum (ZIMSOFF), a national farmer-led network that promotes organic farming, agroecology, and seed sovereignty among smallholder farmers, was central in identifying participants. It was chosen because its women-led seed initiatives and extensive grassroots membership provided both access to experienced seed custodians and a platform for observing collective seed preservation, exchange, and advocacy in practice.

Participant selection

Participants were identified through purposive and snowball sampling. Initial contacts from ZIMSOFF, local farmer groups, and extension officers referred me to seed custodians, community leaders, and members of agroecology collectives. While the sample includes farmers of different genders, ages, and affiliations (including both organized agroecology groups and unaffiliated individuals across four districts), these categories were deliberately included to capture diverse perspectives in seed preservation: gender to reveal differences in roles and recognition; age to explore intergenerational knowledge transmission and adaptation; and affiliation to compare practices between those embedded in formal agroecology networks and those operating independently. These social differences are closely tied to variations in seed access, preservation techniques, and crop choices, all of which directly affect household food security and nutrition. Instead, this study treats the sample as a purposeful case of smallholder farming communities engaging with traditional seed systems under varying levels of exposure to agroecology. The heterogeneity within the participants allows for the comparison of shared and diverging experiences, particularly around gender roles, intergenerational knowledge, and seed system adaptation. The aim is to deepen the empirical illustration and theoretical understanding of how seed practices operate across different social and ecological contexts.

Data collection and analysis

Data collection included 34 semi-structured interviews with 20 women and 14 men aged 25–78, seven FGDs with four mixed-gender and three women-only groups

(Table 1), and participant observation at seed fairs, field days, and household visits. I conducted interviews in English and local languages (IsiNdebele, ChiShona, Kalanga), that I recorded, translated, and transcribed. I scheduled field visits to coincide with key stages of the agricultural calendar. The July–August 2022 visits took place during the post-harvest storage season, allowing observation of preservation and storage techniques for crops such as millet, sorghum, rapoko, and groundnuts; the November–December 2022 visits coincided with land preparation and planting, providing insights into seed selection, planting rituals, and intercropping decisions; and the April–May 2023 visits occurred during the main harvest period, enabling documentation of crop assessment, seed sorting, and preparation for the next cycle. Follow-up visits in late 2023 and throughout 2024 captured adjustments in preservation, exchange, and storage practices in response to climatic variability and shifting resource availability.

I followed Braun and Clarke's [36] thematic method of analysis, combining deductive codes derived from the research objectives and theoretical framework (e.g., gender roles, seed challenges, food security, institutional support) with inductive insights emerging from the data. Deductive coding allowed me to systematically tag segments where participants discussed themes such as women's authority in seed selection, men's roles in hybrid seed adoption, or the impact of climate variability on preservation methods. Inductive coding captured unanticipated patterns, such as local classifications of seeds into ancestral and modern categories, or the role of seed fairs as spaces for both knowledge exchange and social recognition. By iteratively comparing materials across interviews, focus groups, and observations, the analysis identified points of convergence (e.g., widespread use of underground pits for storage) and divergence (e.g., differing valuations of hybrid maize between younger and older farmers), ensuring that the final themes exhibit both shared and context-specific experiences.

The table summarizes the demographic composition of semi-structured interviews and FGDs conducted in Matabeleland, Zimbabwe. Numbers indicate participants by gender, age range, and organisational affiliation. ZIMSOFF refers to the Zimbabwe Smallholder Organic Farmers Forum; 'Independent' denotes farmers without formal organisational membership.

Results

Across all four study communities, participants described a close relationship between seed-saving practices and household resilience, cultural continuity, and autonomy over production. Women in particular framed seed preservation as a way to secure food without dependency on purchased seed, adapt to changing rainfall patterns, and maintain control over what is planted and eaten. These roles were expressed through a variety of agroecological practices, including intercropping, organic composting, and seed exchange, which participants linked to both cultural heritage and practical adaptation strategies.

Cultural significance of seeds

The study revealed that traditional seeds hold profound cultural significance in all four communities (Bubi, Matobo, Nkayi, and Plumtree), serving as central pillars of both agricultural production and community identity. In interviews and focus groups, elderly women regarded seeds as carriers of deep spiritual meaning and cultural heritage. A 65-year-old woman from Matobo who serves as both a seed keeper and community elder in her village, captured this sentiment:

> *Seeds are the pillar of cultivated plants; they represent life itself. Without seeds, there is no food, and without food, there is no life. They are divine creations, the beginning of life and our future. Seeds symbolize independence, strength, and power for us farmers, ensuring the continuity of life.*

Participant observations during seed fairs and household visits revealed how seeds are woven into complex rituals and social protocols with strict gender dynamics. In Nkayi, only pre-menstrual girls were traditionally allowed to handle stored seeds, tied to notions of purity and respect. Gender divisions extended to ceremonial practices, where men often led ritual blessings of seeds (*ukubika inhlanyelo*), a role associated with invoking ancestral protection and community prosperity, while women prepared seed mixtures and maintained preservation knowledge, tasks seen as safeguarding the physical quality and future viability of

Table 1 Demographic and organisational profile of study participants

Data collection method	Number of participants	Age ranges	Females	Males	Affiliation
Semi-structured Interviews	34	25–78	23	11	20 ZIMSOFF affiliated 14 Independent
Focus Group Discussions (FGD)	7 groups 43 participants	25–78	31	12	25 ZIMSOFF affiliated 18 Independent

the seeds. A 73-year-old woman from Bubi remarked on this during an interview, "[t]he men may lead the prayers, but the grandmother knows which ashes to use. Without her, the ritual is not complete."

However, not all participants viewed seed rituals as meaningful. Some younger men prioritized productivity over ritual or symbolic meanings attached to seed practices. Moreover, generational differences emerged in seed perceptions. While elderly women emphasized ritual knowledge and spiritual dimensions, younger participants framed seeds more pragmatically. During an interview in Plumtree, a 28-year-old male agronomist commented:

> *While I respect our traditions, I see seeds primarily as a critical agricultural resource. We need to balance our cultural practices with modern agricultural knowledge to maximize our yields and ensure food security.*

Similar thoughts were endorsed in interviews with some community members expressing concerns about over-relying on traditional varieties in the face of climate change. Some farmers, like a 45-year-old woman in Matobo, appreciated the reliability of certain hybrid varieties in drought conditions. However, she and others still valued traditional seeds for their cultural importance, expressing a desire to uphold their tradition and explore other avenues. As she had experimented with both traditional and hybrid seeds, she stated:

> *Our traditional seeds are important, but we cannot ignore the challenges we are facing. Sometimes, the newer varieties are more resistant to drought or pests and produce more over a short period. We have to find a way to preserve our heritage while also adapting to these new realities.*

Some participants also described local classifications of seed varieties. In Bubi and Nkayi, they distinguished between 'ancestral seeds' (*inhlanyelo yendabuko*) and 'modern seeds' (*inhlanyelo yesimanje*), yet these categories remained fluid and contextual. Not all farmers adhered to rigid categories of 'traditional' or 'modern' seeds. Instead, many described combining both types in their farming systems. This practice was often framed as a practical choice rather than an ideological stance. As one man in his forties, from Nkayi put it, "[w]e grow what works. Sometimes it's the hybrid for a quick harvest, or traditional millet for taste." Farmers often mixed seed sources as adaptive strategies, planting hybrid maize alongside traditional seeds.

Women's roles in seed systems

Across all study communities, women were widely recognized as the main custodians of traditional seed knowledge, responsible for selecting, drying, treating, and storing seeds. Most women learned techniques from their mothers or grandmothers, exemplifying intergenerational knowledge transmission. During a focus group in Matobo, a woman in her sixties explained the depth of this inherited expertise noting, "I learned from my mother how to choose the best cobs and how to use ash to keep them safe from weevils. This is women's knowledge. It is what we pass down." Most women interviewed reported using locally available materials for seed preservation, drawing on traditional ecological knowledge. They employed ash from specific tree species like Combretum imberbe (*umtswiri*) and Aloe ferox (*inhlaba*), goat manure, and underground storage pits. These techniques were directly observed during site visits to Bubi and Matobo, where women demonstrated the use of ash from burnt Combretum imberbe trees to protect seed from pests, confirming the preservation methods described in FGDs.

In addition to seed processing and storage, women also practiced intercropping and organic pest control as part of their agroecological routines. Several participants explained how they plant millet alongside cowpeas or pumpkins to reduce pest pressure and conserve soil moisture. A 63-year-old woman from Plumtree noted in an interview, "[w]e mix crops because they protect each other. When we plant pumpkins under the millet, they cover the soil and keep the ground moist." This was confirmed across all data sources; women described these techniques in interviews, exhibited them during participant observations (e.g., planting pumpkins under millet in Bubi), and discussed them in detail during FGDs. Across these different data sources, the core practices were largely consistent, but the terminology used to describe them varied. Elders tended to use traditional expressions such as 'ancestral planting' while younger women more often adopted contemporary terms like 'climate-smart' or 'water-saving.'

Moreover, women's seed work was strengthened through collective action, particularly in ZIMSOFF-affiliated groups for farmers. FGD participants described exchanging rare varieties, sharing preservation techniques, and co-organizing seed fairs. Women in organized seed networks had greater access to training and new preservation techniques, compared to those not affiliated with any group. Several participants noted that seed fairs and field days gave women opportunities to speak in public and be acknowledged for their seed-saving work. A Bubi group leader in her late sixties explained that during their annual events organized by ZIMSOFF, women

came together from different villages with their seeds and their stories to learn from each other, and leaving with new knowledge and strengthened bonds.

I found shifting gender roles in seed preservation, particularly within mixed-gender agroecology groups where younger men are increasingly participating. While women remain the primary experts in traditional seed knowledge, men are becoming more involved, especially with hybrid seeds, given that younger men often associate themselves with new techniques and modern agricultural innovations. A 35-year-old man from Plumtree stated in an interview; "[t]raditionally, women saved the seeds. But with these new varieties, I have started getting more involved, especially with storage and treatment." Male involvement was most observed among a few younger men exposed to agroecology or climate-smart agriculture training, whereas older men or those outside these networks generally stuck to conventional practices. On that note, some men still distanced themselves from traditional seeds focusing instead on yield. Yet, during household visits and seed events, I rarely observed male participation in seed selection and storage, despite frequent claims made by men during interviews. Women also pointed out this gap, noting that while men increasingly claimed involvement in seed work, especially with newer seed varieties, the daily labour, decision-making, and preservation responsibilities still rested with women.

In FDGs, some women expressed concern that men's growing involvement might overshadow women's long-standing contributions, particularly in public recognition. Observations confirmed that men often received more attention at seed fairs even when women had done most of the preservation and presentation work. However, other women viewed male participation positively, noting it could strengthen family-based seed work and improve knowledge sharing. Most participants described seed-sharing as collaborative, with farmers willingly exchanging techniques and varieties, though some competition over rare seeds was reported. While welcoming shared labor, there was an emphasis ensuring that women's expertise remains acknowledged and leadership stays inclusive.

Structural barriers to traditional seed preservation

Four key categories of barriers to sustaining traditional seed preservation emerged across study communities: economic and infrastructure constraints, policy and institutional gaps, climate variability and genetic contamination, and intergenerational disengagement.

Economic and infrastructure constraints: Many women reported lacking the physical resources required for seed preservation, such as secure granaries, appropriate containers, and consistent labor support. Financial exclusion compounded these material constraints, creating dependency on external seed sources. A 52-year-old widow in Plumtree explained during an interview:

> *Even when we have the knowledge and ability to improve our seed preservation methods, we often lack the resources to implement changes… I know how to store seeds, but I have no proper place. If pests come, I lose my seeds.*

Also, without land titles or collateral, women face significant financial barriers to investing in seed infrastructure making it difficult to access credit. Their seed-saving work also lacks formal recognition, further limiting institutional support. In an FGD, a widow in her fifties, from Kayi, highlighted that banks rarely lend to women, particularly for traditional farming. Meanwhile, reliance on government or NGO-distributed hybrid seeds is growing, creating dependency. She added that while these handouts provide short-term relief, they require annual repurchase, unlike traditional seeds that can be saved and replanted. To support this, A 67-year-old male community leader from Matobo stated during an interview that people prefer waiting for government hybrid seed handouts rather than maintaining their own seed stocks. Many women expressed concern that this shift undermines long-term food security, as traditional seeds offer sustainable, self-sufficient means of food production.

Policy and institutional gaps: Interviews with extension officers from Zimbabwe's Department of Agricultural Technical and Extension Services (AGRITEX) and farmers revealed frustration with agricultural support systems that neglected traditional seed work. A 45-year-old woman from Bubi said, "AGRITEX officers only come to talk about new maize varieties. They do not care how we save our own seeds." This was confirmed by participants in Nkayi and Matobo. Nevertheless, a male extension officer in his late thirties, interviewed in Bubi expressed doubts about the viability of traditional seed systems by saying, "[f]armers want what gives results. Traditional seeds are good culturally, but they are not always reliable. We try to encourage certified seeds because that is what is available and tested." This institutional neglect left women relying on informal networks or temporary donor-funded projects with minimal support for traditional seed storage facilities.

Climate variability and genetic contamination: Farmers reported that climate change has disrupted traditional farming calendars and undermined seed preservation methods. Erratic weather patterns, especially delayed rains and long dry spells, excessive heat, and pest outbreaks were cited as major threats to seed viability and crop performance. A woman in her forties from Nkayi explained during a FDG:

> *Climate change has affected rainfall patterns... Rainfall has become less and less, leading to seeds failing to germinate due to dry ground. On some occasions, the rain is too much, such that the seeds rot underground.*

Field notes from November 2023 documented multiple seed storage failures due to moisture and rot, forcing farmers to turn to hybrid seeds perceived as more adaptable but bringing their own set of challenges. This shift has led to cross-pollination with hybrid varieties posing significant threats to the genetic integrity of traditional seeds. A 36-year-old AGRITEX officer corroborated; "[t]here is now a lot of unintentional cross-pollination… this completely alters the genetic makeup of these seeds. Over time, the traditional seed will lose its character." This erosion of seed purity alarmed some women, who found it increasingly difficult to protect indigenous crop varieties outlined in Table 1.

Intergenerational disengagement: Elderly women frequently expressed concern about youth losing interest in seed work. As a 75-year-old seed custodian from Bubi explained during an interview, "[t]he young people are more interested in quick returns from hybrid seeds. When we try to teach them our methods, they say it is too time-consuming." The concern was important because this knowledge on how to select, store, and protect seeds using local materials is typically transmitted orally and through practice rather than written records, creating the risk that it would be lost permanently when the elderly pass away. However, the picture offered by some younger participants was not entirely bleak. Rather than rejecting traditional practices outright, some (particularly those in agroecology groups) were actively adapting them, experimenting with storage innovations and improved preservation tools to make seed work more compatible with their priorities.

Food sovereignty and nutrition security

Women in all study communities consistently linked seed saving to a sense of autonomy, food diversity, and cultural pride. In interviews, women described traditional seed systems as enabling household food sovereignty: the ability to choose what to grow and eat. Being able to plant without needing to buy seeds each year was seen as a form of independence that provided household benefits. A 67-year-old woman from Matobo explained this, "[w]e plant what we choose because we have our own seeds. We do not wait for anyone. It gives us freedom. But when you depend on seed from shops or government, you are not free." These women pointed out that seed control allowed them to respond to local environmental conditions, personal preferences, and family needs through informal exchanges. However, most participants acknowledged that their production capacity was influenced by the availability of resources such as land, the quantity and quality of their seeds, and prevailing climatic conditions. They also linked agroecological practices to soil improvement and reduced dependence on synthetic inputs. In Plumtree, a focus group of women who are part of ZIMSOFF organization described composting with goat manure and leaf litter to enrich their gardens.

Moreover, in several focus groups, many participants linked traditional crops to specific health benefits. These claims were consistent across interviews, FGDs, and field observations. Women repeatedly described crops like millet and sorghum as "foods that give strength" during household interviews. They substantiated these claims during group discussions and displayed these grains prominently at seed fairs with handwritten nutritional notes. While participants did not reference formal nutritional studies, these perceptions were widespread and consistent across communities. Several women mentioned that health workers had encouraged them to continue growing 'small grains' because of their dietary value (Table 2), lending institutional support to these traditional practices. To illustrate the specific crops that women prioritized for cultivation and preservation, the table below summarizes the key indigenous varieties grown across the study sites, highlighting both their nutritional value and their contribution to food security.

The table shows descriptions that summarize local knowledge of each crop's role in nutrition, resilience, and cultural value. Local names are in isiNdebele, as used by participants.

The crops listed in Table 2 were consistently named by women across all sites as foundational to household food security and nutrition. Participants explained that these crops ensure families have something to eat when maize fails. Women noted that these crops "never fail us," and their ability to be stored for extended periods meant households could draw on them during droughts, late rains, or economic hardship. As one older woman in Bubi stated during a FGD, "[w]ith our seeds, we may be poor, but we do not go hungry." Seed fairs and focus groups in Bubi and Plumtree confirmed this notion. Traditional diets were described as 'more diverse, filling, and real food' compared to commercial maize-based meals. However, some male participants challenged claims that traditional crops were always more nutritious. In an interview, 40-year-old man in Nkayi asserted that, "I do not think it matters that much because maize is still what fills people. These other grains are healthy, yes, but we cannot always grow enough of them." Moreover, some women acknowledged using hybrid maize to cope with short

Table 2 Traditional crops and their contributions to nutrition and food security in Zimbabwe

Crop name	Local name	Nutritional and food security contribution
Sorghum	Amabele	Rich in protein, supports gut health, prevents anemia, and is used for porridge, sadza, and traditional brews. It holds cultural value and provides a reliable food source during maize shortages
Millet	Inyawuthi/Mhunga	Easily digestible, matures quickly, thrives in arid conditions, and serves as a reliable backup when commercial crops fail, making it a key pillar of household food security
Finger millet/Rapoko	Uphoko	Supports bone health and energy, is suitable for diabetics, and stores well as a vital off-season food reserve. Serves as a family buffer against hunger and regarded as an 'ancestral grain.'
Cowpeas	Indumba	High in protein from both seeds and leaves, they serve as grain and vegetable. Their versatility across meals and seasons supports food diversity and resilience, especially in droughts
Groundnuts	Amazambane	Rich in healthy fats and protein, they are eaten roasted or as paste, adding dietary flavour and diversity to meals. Their storability and adaptability support household nutrition and income
Roundnuts	Indlubu/Nyimo	Filling and energy-dense, they grow in poor soils with little rain, making them a reliable survival crop. Women called them a "food of last resort" that "never fail," and a core food security staple in marginal areas
Pumpkin Vegetable	Ithanga	Produces nutrient-rich leaves, fruit, and seeds, a common household relish. Grows with minimal inputs and, when intercropped, boosts dietary diversity and year-round food availability

rains or urgent food needs and some farmers acknowledged a trade-off; hybrid maize had to be bought each season "you cannot replant it like our local sorghum," one man explained, whereas traditional seeds could be saved, albeit with more effort to protect them.

Food sovereignty practices were deeply gendered. While women bore the primary responsibility for seed preservation and meal preparation, their authority over land or income was often limited. Seed work provided pride and recognition but not always power or economic control. Women who were recognized as seed custodians reported both pride and challenges. A 63-year-old seed custodian from Bubi explained this challenge in an interview saying, "[w]e are expected to keep the seeds safe, to know when to plant, and to teach others… sometimes it is hard, especially when the weather changes or when young people do not listen." However, not all women had equal capacity to fulfill seed custodian roles. Some independent seed keepers mentioned that seed work could be isolating or unrecognized unless tied to a formal group or organization.

Drawing on interviews, FGDs, and participant observations, Fig. 1 illustrates how women's seed stewardship spans multiple, overlapping domains, as knowledge holders, food providers, environmental caretakers, and cultural transmitters. As seed guardians, they conserve drought-resistant seed varieties that continue to yield food under challenging conditions. As family providers, they make daily decisions about planting, cooking, and preservation to meet household nutritional needs. As knowledge holders, they ensure that seed preservation, crop rotation, and soil health techniques are passed down and shared across generations.

Illustrates the cyclical relationship between women's roles in seed preservation, food provisioning, and knowledge sharing. Each role reinforces the others: preserving seeds maintains crop diversity, food provisioning ensures access to culturally important foods, and knowledge sharing passes practices across generations.

These practices were closely tied to household autonomy, dietary diversity, and a sense of pride in sustaining traditional crops. While many women emphasized the value of traditional seeds and practices, their experiences were not uniform.

Diverging perspectives and everyday negotiations

A number of participants expressed diverse perspectives on seed choices, shaped by environmental uncertainty, household dynamics, and organizational involvement. While many valued traditional seeds for their role in cultural continuity and autonomy, others pointed to the practical advantages of hybrid varieties during drought or food shortages. A 38-year-old man from Nkayi noted during an interview that hybrid maize was essential for coping with short rains and urgent food needs, and a woman in Plumtree admitted she began planting hybrid maize to ensure food availability for her family, even though she 'appreciated the value' of traditional crops.

Beyond seed preferences, several women raised concerns about how their seed-saving work was recognized or overlooked within households and organizations. These accounts were corroborated by participant observation during seed fairs, where men were often the first to be invited to speak or receive attention, despite women doing most of the seed selection and presentation. In one household visit in Matobo, I observed a similar tension when a male family member dismissed a woman's efforts as "just for show." These dynamics were also reiterated in focus groups, where women voiced frustration about public recognition not translating into material or

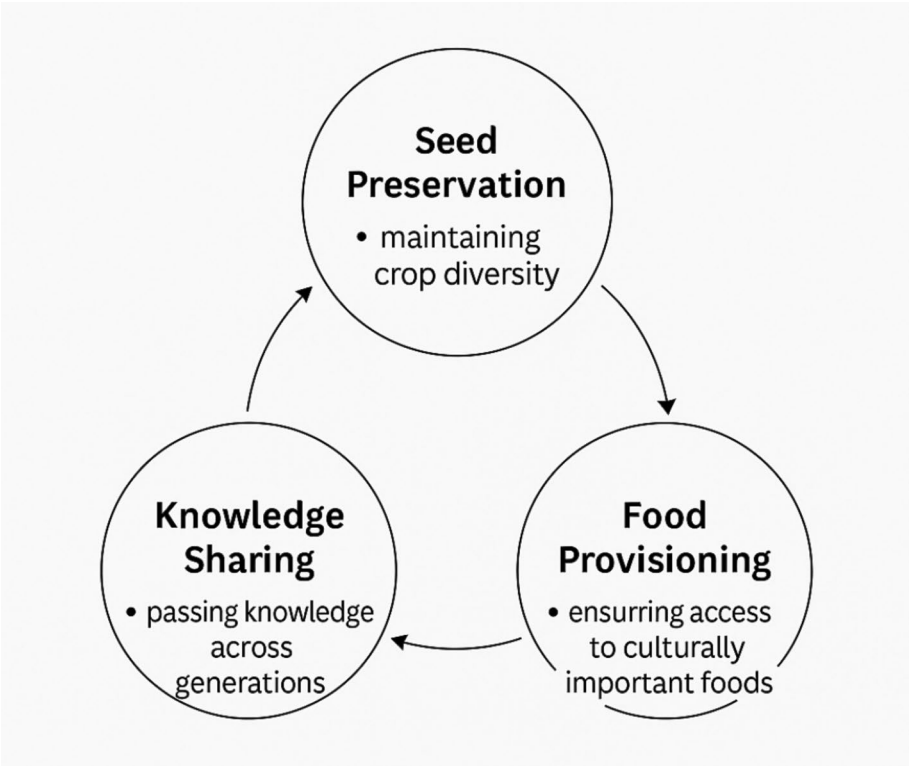

Fig. 1 Women's interconnected roles in traditional seed systems

institutional support. As one woman put it, "[w]e explain everything, but when a man stands up, people really listen." This pattern was evident in different material sources, pointing to persistent gendered asymmetries in both informal and formal seed-related spaces, influencing women's motivation, engagement and ultimately, the strategies they pursued to secure seed access and food security.

The five thematic sections above illustrate how women's roles in traditional seed systems are shaped by cultural values, labour dynamics, institutional gaps, and adaptive strategies. Despite the diversity of lived experiences, consistent patterns emerged across study sites. Table 3 synthesizes these patterns, showing how participants navigated seed system constraints under changing ecological, social, and economic pressures, based on interviews, FDGs, and field observations conducted across four rural communities between 2022 and 2025.

Summarizes key challenges faced by smallholder farmers in maintaining traditional seed systems, the adaptive strategies employed, and the effects reported by participants. Outcomes reveal local observations rather than formal agronomic measurements.

Table 3 Challenges, strategies, and outcomes in Zimbabwean seed systems

Challenge	Adaptive strategy	Observed effect
Poor seed storage	Ash, underground pits	Partial success; losses in extreme weather
Institutional neglect	Informal seed exchange, seed fairs	Sustained practice; low formal recognition
Gendered labor, low recognition	Women's groups, knowledge sharing	Higher visibility locally, but limited authority
Climate stress	Intercropping, mixed seeds	Resilience improved; seed blending normalized
Genetic contamination	Isolating traditional seeds	Some success; varietal purity still at risk
Youth disengagement	Rituals, storytelling, mentorship	Mixed uptake; concern over knowledge loss

Discussion

Reinterpreting traditional seed systems through a feminist agroecological lens

This study confirms that traditional seed systems in Zimbabwe are deeply embedded socio-cultural institutions in which seeds symbolize ancestral continuity, household resilience, and intergenerational knowledge. Women's seed work, far from being a nostalgic remnant, represents a dynamic agroecological and political labor practice. This aligns with findings from Malawi [10] and Tanzania [37], where seed-saving is recognized as both ecological stewardship and socio-political resistance. These practices affirm Santeramo's [38] call to reconceptualize food security beyond technical indicators, integrating cultural values, local agency, and gendered resilience.

In parallel, these findings also reinforce Ncube et al.'s [39] study in eastern Zimbabwe, which showed that local, farmer-managed seed systems, largely driven by women, offer more accessible and culturally grounded seed security than formal commercial channels. These findings illustrate this picture by illuminating the gendered labor and informal governance behind these systems. While Ncube et al. capture the material and logistical dimensions of informal seed systems, this study extends the analysis by highlighting the gendered labor, intergenerational care, and spiritual significance underpinning them. In doing so, it underscores how women's seed practices operate not only as adaptive strategies but also as embodied forms of resistance and cultural continuity that are often rendered invisible in mainstream agricultural policy. Intersecting factors such as gender, climate variability, class, and land tenure actively shape Zimbabwe's seed systems, as revealed through the narratives of women farmers. Similar to Nyantakyi-Frimpong and Bezner Kerr [40] and Naheed and Rukhsana [41], Zimbabwean women's seed practices persist under structural exclusion, while sustaining biocultural knowledge. The generational divergence observed in my study echoes patterns reported by Makate, Makate [42], where younger farmers favor yield and climate resilience, while elders uphold symbolic and spiritual seed meanings.

Significantly, this research challenges the persistent binary between 'traditional' and 'modern' seed systems. Farmers in this study engaged in hybrid strategies, combining ancestral varieties with selected hybrids based on environmental and household needs. This finding mirrors scholars documenting flexible seed practices among African smallholders [5] and in other regions, such as Humphries, Rosas [43] in Honduras, where farmer—scientist partnerships integrated local varieties and farmer priorities into breeding programs. Moreover, Mba, Guimaraes [44] caution that hybridization is neither new nor necessarily disruptive as farmers often adapt modern varieties into local logics. This study extends this by revealing the cultural, spiritual, and gendered motivations behind seed blending, illustrating how seed practices are not just about yield, but about meaning, memory, and survival. These findings also coincide with critiques of technocratic seed system approaches [45], which rely heavily on availability-based indicators and overlook the social, cultural, and gendered dimensions of food systems. Such approaches risk reducing seeds to units of production, ignoring their roles as carriers of heritage, identity, and intergenerational knowledge. This conforms to the work on the multiple values of genetic resources in formal conservation contexts [46], where material traits are often prioritised over cultural and relational meanings.

Gender, power, and institutional contradictions

Although women are widely recognized as custodians of traditional seed knowledge in Zimbabwe, their authority remains structurally marginalized. This reflects Kloppenburg's [47] concept of 'epistemic exclusion,' wherein informal knowledge systems are rendered invisible in formal agricultural policy. Similar dynamics are observed in South Africa [48], where top-down extension services are perceived by farmers as ineffective or misaligned with local realities, often devaluing women's contributions to agroecological resilience. This study supports and extends Agarwal's [49] and Doss's [50] findings on how gendered access to land, credit, and decision-making impedes women's participation in agricultural governance. While ZIMSOFF-affiliated groups provide some institutional recognition, they remain exceptions. Most women remain excluded from leadership roles, despite being at the forefront of seed preservation and agroecological innovation. As summarized in Table 3, despite employing a range of innovative and culturally grounded strategies, women continue to face systemic barriers to institutional recognition and support. Similar gendered participation dynamics are documented by Musara, Bahta [51], where women dominate primary processing in local value chains yet face systemic marginalization in market domains. In this study too, women's expertise in seed systems creates invisible labor that rarely translates into economic or institutional recognition. As noted by Manja, Zingwe [52], intra-household dynamics and land tenure continue to determine women's ability to access improved varieties and resources.

One important aspect is that men's increasing involvement in seed work, especially around hybrids, presents both opportunities for collaboration, it also risks erasing women's historical expertise. These tensions mirror broader patterns in agricultural development, where inclusion strategies can inadvertently recentralize male authority [50]. Women reported similar concerns, with

many women fearing that men's growing participation in seed fairs and farmer groups—while positive in theory—could marginalize their knowledge unless institutional safeguards are implemented. This mirrors findings by Jemaneh and Shibeshi [53], who show how gender-blind innovation programs often re-centralize male authority, even in tasks historically dominated by women.

The Zimbabwean situation exemplifies how current approaches fail to engage with the complex power dynamics shaping seed systems. While agroecology groups like ZIMSOFF have created important spaces for women's leadership, my findings suggest these remain exceptional cases rather than systemic solutions. With this reflection, policy responses must move beyond tokenistic inclusion. This critique demonstrates Tirivangasi et al.'s [54] findings, where Zimbabwean agricultural institutions consistently privileged formal breeding systems while devaluing women's informal seed expertise. His study highlights how top-down seed policies, despite their development goals, often reproduce structural inequalities. My findings extend this by showing how women's seed-saving work is not only invisible to policy actors but also increasingly crowded out by donor-led hybrid seed programs. As Onyeaka, Siyanbola [17] argue, agricultural support systems should be explicitly feminist in design, recognizing seed-saving as skilled labor and prioritizing women's leadership. Without such safeguards, even well-intended agroecological programs risk reproducing the very inequalities they seek to address.

Navigating hybrid systems and strategic adaptation

Farmers in this study operate within complex hybrid systems, a dynamic mix of traditional and commercial seeds. This shows what Almekinders, Beumer [25] describe as the deliberate integration of formal and informal seed systems, using a diverse portfolio of seed types to manage climate risks, market constraints, and dietary needs. These findings resonate with some studies emphasizing smallholder experimentation and localized adaptation [55]. They also conform with Fischer's [56] account of maize in South Africa as both a commodity and an anti-commodity, indicating its dual economic and socio-cultural roles in smallholder systems. This framing concurs with the hybrid strategies observed in Zimbabwe, where farmers blend commercial and indigenous seed varieties not as contradictory choices but as complementary responses to ecological risk, market pressures, and cultural priorities.

My contribution lies in showing how this adaptation is not only ecological, but deeply gendered and symbolic. Women's decisions to preserve ancestral seeds while using hybrids in lean years reflect not contradiction, but coherence, a balancing act between heritage and survival. Table 3 shows this pragmatic balancing act; strategies like seed blending and selective hybrid use emerged not as ideological rejection or acceptance, but as grounded adaptations to risk and climate uncertainty. Sacred seed spaces, communal blessings, and story-based learning practices demonstrate the affective and relational dimensions of agroecological adaptation. Moreover, the study highlights institutional blind spots. AGRITEX and NGO programs often misrecognize hybrid strategies as failures to adopt innovation, rather than deliberate, culturally grounded choices. This verifies Scoones & Thompson's [57] critique of one-size-fits-all seed policies and supports calls for more context-sensitive extension models.

Seed sovereignty as relational empowerment

Seed sovereignty in this context is less about autonomy in the abstract and more about relational empowerment, anchored in land access, communal knowledge, and ritual practice. Women described seed saving as a way to choose what to plant, how to cook, and how to pass knowledge to future generations. These overlapping functions as food providers, environmental stewards, and cultural knowledge holders, are visualized in Fig. 1, which illustrates the interconnected roles women occupy in sustaining seed systems. This finding indicates broader patterns in Zimbabwe's post-land reform context, where women carry the burden of subsistence production but remain excluded from land and decision-making authority [58]. Although the Fast Track Land Reform Programme redistributed land, it did not dismantle entrenched patriarchal hierarchies. My study confirms that women's seed sovereignty is still negotiated within unequal power structures, both in households and institutions. The results endorse the work of Isakson [59], Bezner Kerr, Hickey [22], and Coomes, McGuire [6], who similarly highlight the centrality of seed work to food sovereignty and gender justice.

Women in formal groups gained visibility and sometimes material support; others described being left behind. As in other feminist agroecology studies [29, 60], recognition mattered as much as access. As Fig. 1 suggests, recognition of women's seed labor is inseparable from their daily relational work that spans household nutrition, seed selection, and social continuity. NGOs often required women to reframe seed work using scientific language ('climate-smart') to gain funding, yet many women resisted this, using cultural forms; rituals, kinship ties, or storytelling to assert sovereignty on their own terms. Such dynamics underscore the need for gender-transformative monitoring tools, like the Reach, Benefit, Empower, Transform (RBET) framework proposed by Nchanji, Ageyo [18], which tracks progress from women's inclusion (Reach), to gaining tangible gains (Benefit), to

increased decision-making power (Empower), and ultimately to shifts in norms and institutions that reduce gender inequality (Transform).

In this study, most women were reached through agroecology initiatives and some benefited from training and improved seed access, but fewer reported increased decision-making power, and transformative change remained rare without broader institutional and cultural shifts. This approach emphasizes women's agency and long-term empowerment rather than surface participation in seed programs. These findings expand Onyeaka et al.'s [17] rights-based model by showing how sovereignty is not just claimed, but lived through work, memory, and relational care. The study thus calls for policy shifts that support collective women-led networks, protect land access, and fund bioculturally grounded innovation. My analysis thus aligns with Bezner Kerr's [61] call to reconceptualize food security through feminist agroecological frameworks that center Indigenous and women's knowledge systems.

Food sovereignty and agroecological futures

Traditional seeds offer more than resilience; they embody a future rooted in cultural identity, dietary diversity, and spiritual meaning. Crops like millet, sorghum, and rapoko are important to both nutrition and heritage, particularly during droughts or lean periods. Similar to Dias-Martins, Pessanha [62] and Taylor and Kruger [63], this study finds that these crops are not only climatically resilient but carry embedded knowledge and social value. Nevertheless, participants also acknowledged trade-offs. Some embraced hybrid maize for its short growing season, balancing it with traditional crops to maintain autonomy. This pragmatism challenges romantic narratives of 'pure' agroecology while affirming its pluralistic potential.

This biocultural praxis, however, exists in tension with dominant food security frameworks. Where Cruz-Sánchez et al.'s [45] quantitative model measures caloric availability, the qualitative analysis in this paper exposes the limitations of such metrics in capturing the relational networks that sustain seed systems, from grandmothers teaching seed selection through storytelling to communal seed-blessing ceremonies. As Santeramo [38] argues, methodological choices in constructing composite indicators for food security are never neutral, what is measured depends on the decisions made, and these choices can obscure locally relevant dimensions. This underscores why my findings, rooted in cultural and gendered seed practices, reveal forms of resilience absent from standard indices. This perspective supports Patel's [19] critique of production-focused food security and harmonizes with the pluralistic, justice-oriented agroecology envisioned by Kerr, Madsen [61]. The Zimbabwean case illustrates that agroecology's strength lies not in rejecting science, but in integrating diverse ways of knowing. Feminist agroecological futures must therefore be built on strategic hybridity, gender equity, and the institutional recognition of informal expertise.

Conclusion

Zimbabwe's traditional seed systems are dynamic, gendered spaces where continuity and change are constantly negotiated. Women farmers remain central to the preservation, exchange, and adaptation of indigenous seeds, practices that sustain food sovereignty, biodiversity, and cultural identity. However, these contributions occur under intensifying pressures; climate variability, market-led promotion of hybrid seeds, and persistent gendered inequalities in access to land, credit, and institutional support. Rather than romanticizing traditional practices or rejecting modern inputs, farmers, women and increasingly men are crafting hybrid seed strategies that blend cultural knowledge with adaptive innovation. This pragmatic pluralism challenges binary framings of 'traditional' versus 'modern' agriculture and reveals agroecology as a negotiated, context-dependent practice. The emergence of male participation in seed work, especially among younger farmers, marks a notable shift in gender dynamics. While this may enhance recognition of seed practices, it also risks reinforcing institutional bias unless women's expertise continues to be foregrounded and supported. Importantly, the study illustrates that women's empowerment within seed systems is neither automatic nor uniform. It is shaped by intersecting factors such as marital status, age, class, and organizational affiliation. Some women gain visibility and voice through agroecology groups; others remain peripheral despite their active roles in seed work. Recognizing these internal differences is essential to designing interventions that are context-specific, inclusive, and co-designed with women farmers themselves.

In closing, this research highlights the need for policies and programs that go beyond technical solutions to support the social infrastructure of seed systems. Ensuring equitable access to resources, institutional recognition of informal knowledge, and gender-sensitive support mechanisms is critical. Crucially, this also requires embracing seed sovereignty not only as a right to save and exchange seeds but as a form of relational empowerment, rooted in women's everyday practices of care, cultural transmission, and environmental stewardship. These seed practices are directly tied to household food security and nutritional diversity, particularly in drought-prone and economically marginal settings. To make this possible, policy interventions should prioritize support for

women-led seed fairs, protect communal land tenure systems, and legally recognize traditional seed exchanges as legitimate contributions to national food systems. By acknowledging the complexity, hybridity, and gendered negotiations within traditional seed systems, we can foster agroecological futures that are not only more sustainable, but also more food-secure, just, and inclusive.

Abbreviations

AGRITEX	Agricultural Technical and Extension Services (Zimbabwe)
FGD	Focus Group Discussion
FIES	Food Insecurity Experience Scale
FOMPO	Farmer Organization Models for Prospering Smallholder Farmers
FPE	Feminist Political Ecology
FTLRP	Fast Track Land Reform Programme
NGO	Non-Governmental Organization
RBET	Reach, Benefit, Empower, Transform (framework)
SSA	Sub-Saharan Africa
ZIMSOFF	Zimbabwe Smallholder Organic Farmers Forum

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Data availability
The datasets analysed during the current study are not publicly available due to the possibility of linking individuals to specific interviews but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This research, conducted as part of a doctoral degree at Lund University, complies with ethical principles, good research practice guidelines, and legal requirements relevant in Sweden, as well as the ethical standards mandated in Zimbabwe. Although the Swedish Ethical Review Authority did not require separate approval for this specific Ph.D. study, ethical clearance was obtained through the larger FORMAS-funded Farmer Organization Models for Prospering Smallholder Farmers (FOMPO) project (Grant No. 2020-00397). Permission to conduct the research was granted by the Zimbabwe Smallholder Organic Farmers Forum (ZIMSOFF), local authorities, and traditional leaders.

Competing interests
The authors declare no competing interests.

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